Note

Given any three vectors $\vec{a},\vec{b}$ and $\vec{c}$, the following are scalar triple products:

$(\vec{a}\times\vec{b})\cdot\vec{c}$, $(\vec{b}\times\vec{c})\cdot\vec{a}$, $(\vec{c}\times\vec{a})\cdot\vec{b}$, $\vec{a}\cdot(\vec{b}\times\vec{c})$, $\vec{b}\cdot(\vec{c}\times\vec{a})$, $\vec{c}\cdot(\vec{a}\times\vec{b})$,

$(\vec{b}\times\vec{a})\cdot\vec{c}$, $(\vec{c}\times\vec{b})\cdot\vec{a}$, $(\vec{a}\times\vec{c})\cdot\vec{b}$, $\vec{a}\cdot(\vec{c}\times\vec{b})$, $\vec{b}\cdot(\vec{a}\times\vec{c})$, $\vec{c}\cdot(\vec{b}\times\vec{a})$

Geometrical interpretation of scalar triple product

Geometrically, the absolute value of the scalar triple product $(\vec{a}\times\vec{b})\cdot\vec{c}$ is the volume of the parallelepiped formed by using the three vectors $\vec{a},\vec{b}$, and $\vec{c}$ as co-terminus edges. Indeed, the magnitude of the vector $(\vec{a}\times\vec{b})$ is the area of the parallelogram formed by using $\vec{a}$ and $\vec{b}$; and the direction of the vector $(\vec{a}\times\vec{b})$ is perpendicular to the plane parallel to both $\vec{a}$ and $\vec{b}$.

Therefore, $|(\vec{a}\times\vec{b})\cdot\vec{c}|$ is $|\vec{a}\times\vec{b}||\vec{c}||\cos\theta|$, where θ is the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ .From Fig. 6.17, we observe that $|\vec{c}||\cos\theta|$ is the height of the parallelepiped formed by using the three vectors as adjacent vectors. Thus, $|(\vec{a}\times\vec{b})\cdot\vec{c}|$ is the volume of the parallelepiped.

The following theorem is useful for computing scalar triple products.

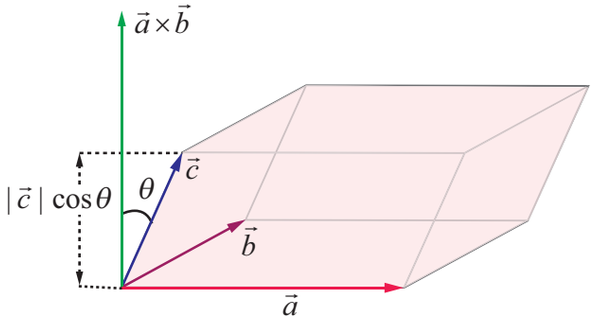


Fig. 6.17

Theorem 6.1

If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$, then

$$(\vec{a}\times\vec{b})\cdot\vec{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

Proof

By definition, we have

$$(\vec{a}\times\vec{b})\cdot\vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}\cdot\vec{c}$$

$$= \left[(a_2b_3-a_3b_2)\hat{i}-(a_1b_3-a_3b_1)\hat{j}+(a_1b_2-a_2b_1)\hat{k}\right]\cdot(c_1\hat{i}+c_2\hat{j}+c_3\hat{k})$$

$$= (a_2b_3-a_3b_2)c_1+(a_3b_1-a_1b_3)c_2+(a_1b_2-a_2b_1)c_3$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

which completes the proof of the theorem.

6.4.1 Properties of the scalar triple product

Theorem 6.2

For any three vectors $\vec{a},\vec{b}$, and $\vec{c}$, $(\vec{a}\times\vec{b})\cdot\vec{c} = \vec{a}\cdot(\vec{b}\times\vec{c})$.

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